computing. Cloud architecture: Cloud scheduling, Scalability, reliability and security of the cloud, Workflow management in cloud, Network infrastructure for cloud computing, Virtualization technologies and its security related issues; Cloud service Models: Software as a Service (SaaS), Platform as a Service (PaaS), google AppEngine, Microsoft Azure etc, Infrastructure as a Service (IaaS), Openstack, EC2 etc, Data as a Service (DaaS); Cloud computing applications: Virtual private cloud , Scientific services and data management in cloud, Enterprise cloud, Medical information systems; Big Data Introduction: Variety of Data, Velocity of Data, Veracity of Data, Distributed file system such as Hadoop, Data centric computing such as map-reduce, Distributed database. Cloud business models.

Recommended Texts:

1. B. Furht and A. Escalante, *Handbook of Cloud Computing*, 1st ed. Springer, 2010
2. R. Buyya and C. Vecchiola, *Mastering Cloud Computing: Foundations and Applications Programming*, 1st ed. Morgan Kaufmann, 2013

CSE 4561 Digital Image Processing Credit: 3.00

Introduction to Signal Processing, Pattern Processing, Computer Graphics, Artificial Intelligence, Human Visual System, Digital Image Representation : Acquision, Storage & Display, Sampling and Quantization, Uniform and Non-uniform Sampling Image Geometry : Perspective Transformation, Synthetic Camera Approach, Stereo Imaging, Image Transform : FFT, PFT, Sine Transformation, Cosine Transformation, Image Enhancement : Spatial and Frequency Domain, Smoothing and Sharpening, Edge Detection, Histogram : Grey Level, Binary Image, Thresh Holding, Half-toning, Image Segmentation : Mathematical Morphology, Dilation and Erosion, Opening and Closing, Image Restoration : Gradation Model, Constrain and Unconstraint Restoration, Inverse Filtering, Wieners Filtering, Image Compression: Source Coding-decoding, Channel Coding-decoding, Practical Image Processing : Electronic Formation of Images, Speed / Memory Problem, Architectures, Decompositions and Algorithms, Computer Implementations for Image Processing Task.

Recommended Texts:

1. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2017
2. M. Sonka et al., *Image Processing, Analysis, and Machine Vision*, 4th ed. Cengage Learning, 2014
3. T. Morris, *Computer Vision and Image Processing*. Red Globe Press, 2003

SWE 4539 Integrated Software Development Credit: 3.00

Distributed version control systems, unit testing and integration testing, automated building, automated packaging, automated releasing, performance monitoring, performance testing, performance tuning, scaling large application, vertical and horizontal scaling, evolutionary database design, automated deployment of production database, continuous integration, continuous testing, continuous delivery, issue tracking, software documentation.

CSE 4554 Machine Learning Lab Credit: 0.75

Sessional works based on CSE 4553.

CSE 4556 Data Mining Lab Credit: 0.75

Sessional works based on CSE 4555.

CSE 4558 Pattern Recognition Lab Credit: 0.75

Sessional works based on CSE 4557.

CSE 4560 Introduction to Cloud Computing Lab Credit: 0.75

Sessional works based on CSE 4559.

CSE 4562 Digital Image Processing Lab Credit: 0.75

Sessional works based on CSE 4561.

SWE 4540 Integrated Software Development Lab Credit: 0.75

Sessional works based on SWE 4539.

The tools and technologies used in the lab may include Git Distributed Version Control System, Jenkins/TeamCity/Bamboo for Continuous Integration and Continuous Testing, Octopus Deploy for automated release management, Jira for Issue Tracking, Confluence for Documentation, Markdown and Wiki Syntax.